Introduction

Customer churn occurs when customers or subscribers stop doing business with a company or service, also known as customer attrition. It is also referred as loss of clients or customers. One industry in which churn rates are particularly useful is the telecommunications industry, because most customers have multiple options from which to choose within a geographic location.

Problem Statement

With the increasing competition in the streaming service market, accurately identifying which customers are likely to cancel their subscriptions ("churn") is crucial for businesses to proactively retain customers and optimize pricing strategies. However, traditional methods of churn prediction often rely on limited customer data, leading to inaccurate predictions.

Customer Churn Analysis Importance for Telecom Industry

There are multiple areas this analysis is so important for telecom industry for future growth and the pros and cons to consider for customers and streaming services and subscription verities of customer required based on demographics and location etc.

Here are the 3 important aspects

Revenue Impact:

Churn directly affects a streaming service's revenue, as lost subscribers represent lost income.

Targeted Marketing:

By identifying at-risk customers, companies can implement targeted retention campaigns, offering personalized incentives to encourage continued subscriptions.

Improved Content Strategy:

Analyzing churn patterns can provide valuable insights into what content types or viewing habits are most likely to lead to customer loss, allowing for better content curation and development.

Why Telecom Industry a Data Science Problem

Telecom Industry Churn Analysis is considered a classic data science problem, specifically falling under the category of "classification" where the goal is to predict whether a customer is likely to churn (leave the service) based on their usage patterns and other relevant data, allowing companies to proactively take steps to retain them; this typically involves using machine learning algorithms to analyze customer data and identify patterns that indicate churn risk.

Data science plays a crucial role in churn analysis in telecom. Data scientists use a range of techniques, including machine learning and predictive analytics, to analyze customer data and develop models that predict which customers are at risk of churning.

Here are few data points detailed out -

Large Data Sets:

Streaming services collect vast amounts of user data including viewing history, time spent watching, device usage, genre preferences, and demographic information, which can be analyzed using machine learning algorithms to identify patterns associated with churn.

Feature Engineering:

Extracting meaningful features from raw data, such as calculating "days since last viewing" or "number of unique titles watched" is critical for building accurate predictive models.

Model Selection:

Choosing the appropriate machine learning algorithms (e.g., Logistic Regression, Random Forest, Gradient Boosting) based on the data characteristics and desired prediction accuracy is essential.

Evaluation Metrics:

Metrics like precision, recall, and AUC-ROC are used to assess the performance of churn prediction models, allowing for comparison and optimization.

R Packages required for this project

- library(plyr)
- library(corrplot)
- library(ggplot2)
- library(gridExtra)
- library(ggthemes)
- library(caret)
- library(MASS)
- library(randomForest)
- library(party)

Following are the datasets required for this project and will move forward with these data sets including few more once started making the EDA on it.

https://www.kaggle.com/datasets/palashfendarkar/wa-fnusec-telcocustomerchurn
https://community.ibm.com/community/user/my-community

Plot and graphs for EDA

To create plots and tables, primarily use the plot () function for basic visualizations and the table () function to generate frequency tables, with the popular "ggplot2" package offering more advanced customization options for creating visually appealing plots.

Basic Plotting will be used in EDA

Scatterplot

This creates a scatter plot where points are plotted based on the values in vectors "x" and "y".

Histogram

Generates a histogram showing the distribution of values in the "variable" column of your data frame "data".

Barplot

Creates a bar chart where the height of each bar represents the count of each unique category in the "category" column.

Boxplot

Displays boxplots to compare the distribution of "value" across different "group" categories.

Question for future steps

Define Churn Rate over term

Monthly charges impact on churn rate

Churn rate based on the cable type provisioned by different customers

Few Statistical Modeling questions